

Chitra Kumar Sai Chenuri Venkata

chitrakumarsai@gmail.com | 316-559-7502 | [GitHub](#) | [LinkedIn](#) | Visa F1 – CPT | [Portfolio](#)

SUMMARY

Data Scientist with 4+ years of experience leveraging advanced analytics and machine learning to optimize business outcomes. Previously, I led a project that improved operational efficiency by 25%, resulting in \$2.8M annual savings. Also, I am [AWS Certified Machine Learning Specialist](#) with a record of accomplishment of delivering actionable insights that drive strategic decision-making.

TECHINICAL SKILLS

Languages: Python, R, SQL, PyTorch/TensorFlow, PySpark.
ML & Deep Learning: Scikit-learn - regressions/classification - SVM, GLM, boosting, random forest, CNN, RNN, LSTM, Transformers
Statistical: A/B testing (z-test, t-test), probability.
LLM/GenAI: Azure OpenAI, LangChain, LangGraph, LLamaindex, LoRA, inspect_ai (LLM evaluation).
Tracking & Deployment: Snowflake-MLOps, Docker, Databricks, MLflow, Onnx.

PROFESSIONAL EXPERIENCE

- Data Scientist | Koch (INVISTA) | Wichita, KS, USA | Manufacturing** May 2022 – Present
- Enabled \$1.2M in strategic IP development by building a patent intelligence system using Azure OpenAI agents to automatically identify at-risk and relevant patents, reducing manual review time by 60%.
 - Drove \$3M in design optimization decisions by automating material composition analysis for automobiles using Azure OpenAI and example-based image prompts, cutting manual effort by 40%.
 - Reduced inference costs by 45% and enabled fraud detection by fine-tuning Azure OpenAI models on telemetry data using LoRA and quantization techniques, improving latency and model efficiency.
 - Projected to unlock \$2M+ in cross-functional value by designing the PRAISE framework (Predictive Reasoning with Actionable Insights and Structured Explanations) to deliver interpretable ML outputs and accelerate AI adoption.
 - Increased customer self-service conversion by 25%, as validated through A/B testing, by optimizing ad placement and embedding top-selling Amazon product listings into the INVISTA website.
- Data Engineer | Fractal Analytics | Remote, USA | Consulting/Marketing** Oct 2021- April 2022
- Improved landing page conversion rate by 22%, through dynamic content personalization powered by clustering and NLP-based segmentation, after training ML models on user interaction and engagement data.
 - Improved advertiser satisfaction by 42% by developing a Looker Studio dashboard for Google Ads to monitor Repair Turnaround Time for 550+ global users, using A/B-tested insights to enhance service workflows.
- Junior Software Engineer | Mindtree Ltd | Hyderabad, TS, India | Consulting/Operations.** July 2017 – June 2019
- Saved \$1M annually by building Python-based data ingestion pipelines from SQL, Azure Data Lake to create pricing analysis views that flagged anomalies and improved price accuracy.
 - Improved customer utilization ratio by 25% by developing a clustering model using Azure ML to detect and rank fraudulent customers via Azure billing and subscription APIs.
 - Generated \$11M in client revenue by enhancing a pet product recommendation system using NLP to analyze consumer reviews and deliver personalized business insights.
 - Boosted customer satisfaction score from 3.8 to 4.5 by leading a 5-member team to build a sales forecasting model using XGBoost, achieving a MAPE below 4.4%.

CURRENT PROJECTS

- Uber ETA Predictions** - Developed machine learning models to accurately estimate arrival times using historical ride data.
Demand Forecasting: Predicted power consumption by analyzing historical data with advanced machine learning techniques.
YouTube/Speech to Text: Implemented transcription & translation services for YouTube videos & audio files - OpenAI Whisper.
RAG App: Created a full-stack application for document upload, indexed querying, and LLM-powered responses.

EDUCATION

- Doctor's Degree, Business Administration and Management (Quantitative Research Focus)** 3.78 | Expected June 2027
Ottawa University, Ottawa, Kansas, USA.
Research Focus: AI-driven strategy execution and organizational intelligence
- Master of science in Computer Science** 3.97 | May 2021
Wichita State University, Wichita, Kansas, USA.

PUBLICATIONS OR AWARDS

- Automated Patent Monitoring Using Azure OpenAI for Strategic IP Protection – Internal Whitepaper, INVISTA R&D, 2023
- Finalist – [2025 Manufacturing Leadership Award](#), National Association of Manufacturers, recognized for developing the PRAISE framework to drive interpretable AI adoption and operational efficiency at INVISTA.
- 1st place on 0 resets & 4th for quickest track completion in all Koch - AWS DeepRacer reinforcement learning competition